

FRANCESCO GANIS

PhD, Sound & Music Engineer

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francescoganis.github.io

Francesco Ganis

Copenhagen, DK



With a strong background in audio digital signal processing (DSP), prototyping, and the evaluation of audio solutions, I bring experience in developing user-centered audio technologies and vibrotactile feedback systems. My work has involved translating user and system needs into structured designs, defining experimental approaches, and iterating on prototypes based on data and user feedback. I am a passionate audio engineer and researcher, with deep love for music and sound. I am eager to contribute, while continuing to expand my knowledge and skills. You can find my portfolio at francescoganis.github.io.

EDUCATION

PhD, Music Training Tools for Children with Hearing Impairment

Aalborg University

Sep. 2022 – Mar. 2026

Copenhagen, DK

Skills learned Research Prototyping Evaluating

- Translated user needs from children with hearing impairments and clinicians into system requirements and prototypes
- Designed experimental methodologies to assess system performance and user interaction
- Developed and iterated prototypes using data-driven and user feedback insights
- Collaborated with audiologists and clinicians to align solutions with clinical practice
- Produced scientific and technical documentation of system behavior, experiments, and results

Master's Degree, Sound & Music Computing

Aalborg University

Sep. 2020 – Jun. 2022

Copenhagen, DK

Skills learned DSP Prototyping Evaluating

- Thesis: "Vibrotactile Feedback for Musical Training of Cochlear Implants Users"
- Completed courses in Sound Processing, Embodied Interaction, Music Perception and Cognition, New Interfaces for Musical Expression

Bachelor's Degree, Electronic Music & Sound Engineering

"C. Pollini" Conservatory

Sep. 2016 – Apr. 2020

Padova, IT

Skills learned Sound Engineering MAX/MSP Sound Design

- Graduated with honors
- Thesis: "Continuous Sonification of Autonomous Driving Vehicles with Purposes of Passengers Information and Trust Augmentation"

STRENGTHS

Technical Skills

MATLAB Prototyping DSP
Product Evaluation Audio Testing
C Arduino Git Unity C#
Embedded Programming ESP32
LaTeX Lasercutting MAX/MSP
Pro Tools Pure Data Reaper
Sound Engineering 3D Printing

Soft Skills

Analytical Thinking Collaboration
Detail-oriented Problem-solving
Scientific Dissemination Supervision
Teamwork Technical Documentation

OTHER EXPERTISE

Photography Playing Drums

LANGUAGES

English ● ● ● ● ●
Full professional proficiency

Italian ● ● ● ● ●
Mother tongue

PUBLICATIONS

Journal Articles

- F. Ganis et al., "The role of haptics in training and games for hearing-impaired individuals: A systematic review," *Multi-modal Technologies and Interaction*, vol. 8, no. 1, p. 1, 2023.

EXPERIENCE

Internship, Technology Evaluation Engineer

Bang & Olufsen

Apr. 2026 – May 2026

Lyngby, DK

Skills learned Product Evaluation Perceptual Testing

Technical Documentation

- Development of experimental paradigm for the perceptual evaluation of high-end audio products
- Definition of test setups and evaluation criteria
- Communication of results and documentation of testing procedures

Teaching Assistant, Sound & Music Computing

Aalborg University

Sep. 2022 – Aug. 2025

Copenhagen, DK

Skills learned Mentoring Communication Supervision

- Assisting students in the *Sound Processing* course
- Assisting students in the *Physical Models for Sound Synthesis* course
- Supervision of student projects in Medialogy and Sound & Music Computing programs

Student Helper, Sound Synthesis and Design

Aalborg University

Oct. 2021 – Apr. 2022

Copenhagen, DK

Skills learned Audio Programming DSP Web Development

- Created sound examples in MATLAB for the book *Serafin, S., Buxton, B., Gaver, B., & Bly, S. (2022). Auditory interfaces. Focal Press*
- Developed and maintained websites

Internship, Prototype Development

Oticon Medical

Oct. 2021 – Jan. 2022

Copenhagen, DK

Skills learned Product Development Prototyping Evaluation

- Development of smartphone cover that provides high-quality vibrotactile feedback to support music listening for individuals with hearing impairment

Sound Engineer, Studio and Live Sound Engineering

Various

2017 – 2018

Veneto Region, IT

Skills learned Sound Engineering Live Electronics

- Centro D'Arte & C. Pollini Conservatory - *Live Electronics*
- RmS Recording Studio - *Boom Operator, Editing Technician*
- Bluetrain's Studio - *FoH Sound Engineer, Live Stage Technician*

Internship, Industrial Prototyping

Danieli Automation S.p.A.

Jun. 2013 – Jul. 2013

Buttrio, IT

Skills learned Prototyping Industrial Design

- Design and creation of a working prototype of a measurement system for foundry equipment

Conference Proceedings

- F. Ganis et al., "Vibrotactile teddy bear: Enhancing musical experiences for children with cochlear nerve deficiency through a vibrotactile soft toy," *SMC Proceedings*, 2025, pp. 115–120.
- F. Ganis et al., "Vibrotactile memory: A case study of timbre perception training in children with cochlear implants using a video game," *SMC Proceedings*, 2024, pp. 134–141.
- F. Ganis et al., "Tickle tuner-haptic smart-phone cover for cochlear implant users' musical training," in *HAID*, Springer Nature, 2022, p. 14.
- F. Ganis et al., "Real-time timbre transfer and sound synthesis using ddsdp," *SMC Proceedings*, 2021, pp. 175–182.

Albums

- M. Finco et al., *Six flute sonatas op.1*, album, EAN Code: 7.93588412371, Japan, 2018.

FURTHER TRAINING

Masterclass, Live Electronics

Chigiana Academy

Jul. 2019

Siena, IT

Masterclass, Symphonic Orchestra Recording

G. A. Fano Institute

Jul. 2018

Spilimbergo, IT

Diploma, Music, Theory and Solfège

G. Tartini Conservatory

Sep. 2009

Trieste, IT

REFERENCES

Prof. Stefania Serafin

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Prof. Lone Percy-Smith

@ Righospitalet, Copenhagen - DK

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Prof. Nicola Bernardini

@ Conservatorio di Latina, Latina - IT

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